FINITELY MANY RATIONAL POINTS OF A LOCALLY ALGEBRAIC GROUP SCHEME LAND IN AN AFFINE OPEN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Proposition proposition-connected-component (canonical tag 0B7R). cited at proof lines 1558-1560. Supplies the irreducible identity component G^0.

Source reference. Original proposition 0B7R: sources/stacks-chow/tags.txt, source lines 14114–14114. Cited at proof lines 1558-1560 for the identity component and its irreducibility used below; load-bearing.

Lemma 0.2. *Let k be an algebraically closed field. Let G be a locally algebraic group scheme over k. Let $g_1, \ldots, g_n \in G(k)$ be k-rational points. Then there exists an affine open $U \subset G$ containing $g_1, \ldots, g_n$.*

Proof. We first argue by induction on n that we may assume all g_i are on the same connected component of G. Namely, if not, then we can find a decomposition $G = W_1 \amalg W_2$ with W_i open in G and (after possibly renumbering) $g_1, \ldots, g_r \in W_1$ and $g_{r+1}, \ldots, g_n \in W_2$ for some $0 < r < n$. By induction we can find affine opens U_1 and U_2 of G with $g_1, \ldots, g_r \in U_1$ and $g_{r+1}, \ldots, g_n \in U_2$. Then

$$g_1, \ldots, g_n \in (U_1 \cap W_1) \cup (U_2 \cap W_2)$$

is a solution to the problem. Thus we may assume $g_1, \ldots, g_n$ are all on the same connected component of G. Translating by g_1^{-1} we may assume $g_1, \ldots, g_n \in G^0$ where $G^0 \subset G$ is as in Proposition 0B7R. Choose an affine open neighbourhood U of e, in particular $U \cap G^0$ is nonempty. Since G^0 is irreducible we see that

$$G^0 \cap (Ug_1^{-1} \cap \ldots \cap Ug_n^{-1})$$

is nonempty. Since $G \to \mathrm{Spec}(k)$ is locally of finite type, also $G^0 \to \mathrm{Spec}(k)$ is locally of finite type, hence any nonempty open has a k-rational point. Thus we can pick $g \in G^0(k)$ with $g \in Ug_i^{-1}$ for all i. Then $g_i \in g^{-1}U$ for all i and $g^{-1}U$ is the affine open we were looking for. □

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